# QUASI-COMPACTNESS OF A MORPHISM OF ALGEBRAIC STACKS CAN BE CHECKED AFTER A FLAT COVER OF THE TARGET

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*Allowed standard prerequisites (editorial).*

- The definition of a quasi-compact morphism of algebraic stacks through the quasi-compactness of the base changes along morphisms from quasi-compact algebraic stacks. The author's section on quasi-compact morphisms, in which the counted statement occurs. Formulates what has to be proved after the given base change is known to be quasi-compact.
- The existence of a smooth surjection from a scheme onto an algebraic stack, and the possibility of shrinking a flat surjective cover of a quasi-compact stack to a quasi-compact open. Used inside the counted proof (source lines 2165-2171). Produces a quasi-compact scheme cover of the test stack through which the quasi-compactness can be transported.

**Lemma 0.1.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $W \to \mathcal{Y}$ be surjective, flat, and locally of finite presentation where $W$ is an algebraic space. If the base change $W \times_{\mathcal{Y}} \mathcal{X} \to W$ is quasi-compact, then $f$ is quasi-compact.*

*Proof.* Assume $W \times_{\mathcal{Y}} \mathcal{X} \to W$ is quasi-compact. Let $\mathcal{Z} \to \mathcal{Y}$ be a morphism with $\mathcal{Z}$ a quasi-compact algebraic stack. Choose a scheme $U$ and a surjective smooth morphism $U \to W \times_{\mathcal{Y}} \mathcal{Z}$. Since $U \to \mathcal{Z}$ is flat, surjective, and locally of finite presentation and $\mathcal{Z}$ is quasi-compact, we can find a quasi-compact open subscheme $U' \subset U$ such that $U' \to \mathcal{Z}$ is surjective. Then $U' \times_{\mathcal{Y}} \mathcal{X} = U' \times_W (W \times_{\mathcal{Y}} \mathcal{X})$ is quasi-compact by assumption and surjects onto $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}$. Hence $\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}$ is quasi-compact as desired. □

*Date*: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.